

```
1 import zipfile
2
3 with zipfile.ZipFile("titanic.zip", 'r') as zip_ref:
4     zip_ref.extractall("titanic")
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Info:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column      Non-Null Count  Dtype
---  ---
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
10   Cabin        204 non-null    object
11   Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None
```

Summary Statistics:

	PassengerId	Survived	Pclass	Name	Sex	\
count	891.000000	891.000000	891.000000	891	891	
unique	NaN	NaN	NaN	891	2	
top	NaN	NaN	NaN	Dooley, Mr. Patrick	male	
freq	NaN	NaN	NaN	1	577	
mean	446.000000	0.383838	2.308642	NaN	NaN	
std	257.353842	0.486592	0.836071	NaN	NaN	
min	1.000000	0.000000	1.000000	NaN	NaN	
25%	223.500000	0.000000	2.000000	NaN	NaN	
50%	446.000000	0.000000	3.000000	NaN	NaN	
75%	668.500000	1.000000	3.000000	NaN	NaN	
max	891.000000	1.000000	3.000000	NaN	NaN	

	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
count	714.000000	891.000000	891.000000	891	891.000000	204	889
unique	NaN	NaN	NaN	681	NaN	147	3
top	NaN	NaN	NaN	347082	NaN	G6	S
freq	NaN	NaN	NaN	7	NaN	4	644
mean	29.699118	0.523008	0.381594	NaN	32.204208	NaN	NaN
std	14.526497	1.102743	0.806057	NaN	49.693429	NaN	NaN
min	0.420000	0.000000	0.000000	NaN	0.000000	NaN	NaN
25%	20.125000	0.000000	0.000000	NaN	7.910400	NaN	NaN

50%	28.000000	0.000000	0.000000	NaN	14.454200	NaN	NaN
75%	38.000000	1.000000	0.000000	NaN	31.000000	NaN	NaN
max	80.000000	8.000000	6.000000	NaN	512.329200	NaN	NaN

```

1 # 4. Value Counts for Categorical Features
2 print("\nSex:\n", train_df['Sex'].value_counts())
3 print("\nEmbarked:\n", train_df['Embarked'].value_counts())
4 print("\nPclass:\n", train_df['Pclass'].value_counts())

```



```

Sex:
Sex
male      577
female    314
Name: count, dtype: int64

```

```

Embarked:
Embarked
S      644
C     168
Q      77
Name: count, dtype: int64

```

```

Pclass:
Pclass
3     491
1     216
2     184
Name: count, dtype: int64

```

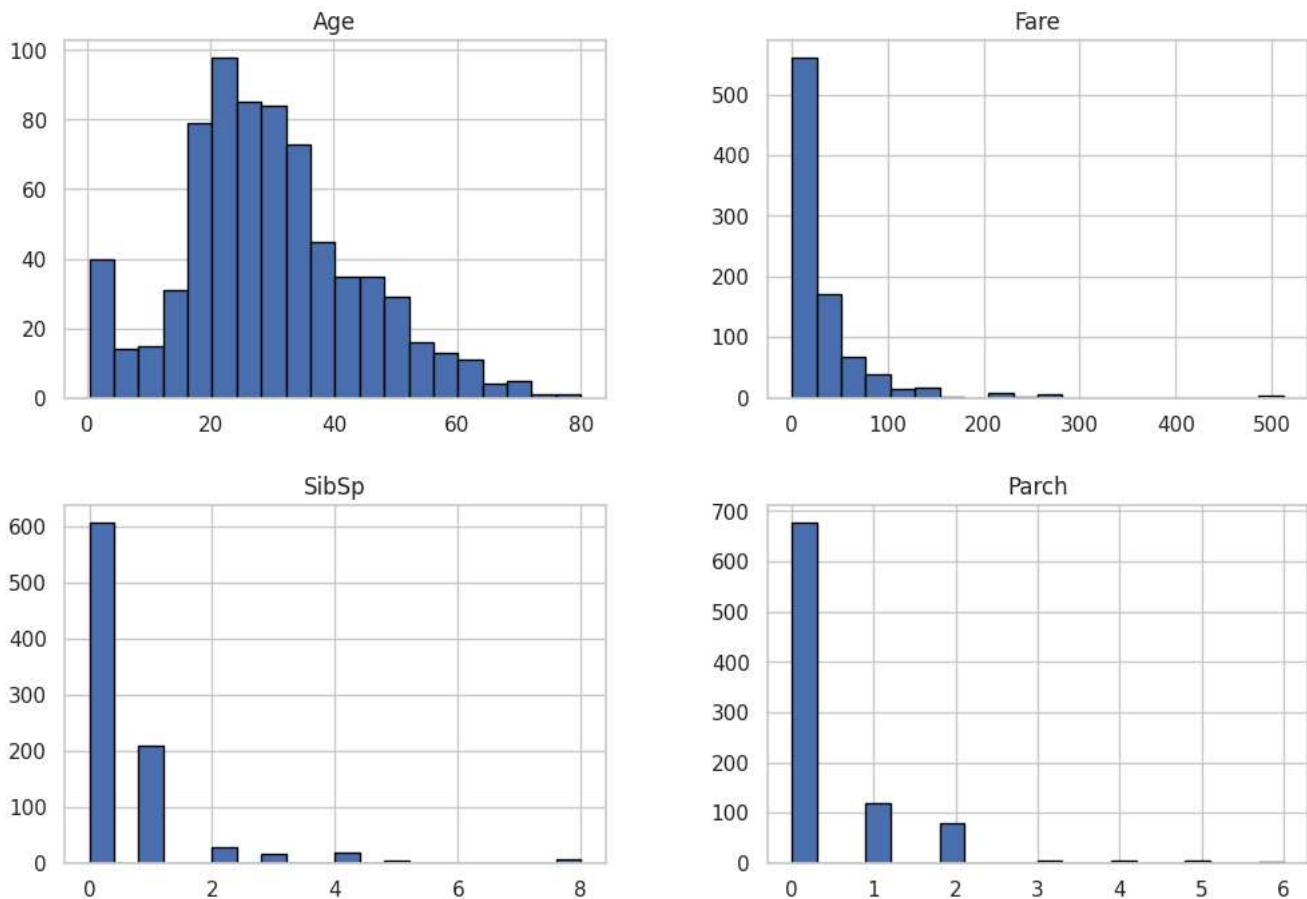
```

1 # 5. Histograms for Numerical Features
2 train_df[['Age', 'Fare', 'SibSp', 'Parch']].hist(bins=20, figsize=(12, 8), edgecolor='black')
3 plt.suptitle("Histograms of Numerical Features")
4 plt.show()

```



Histograms of Numerical Features



```

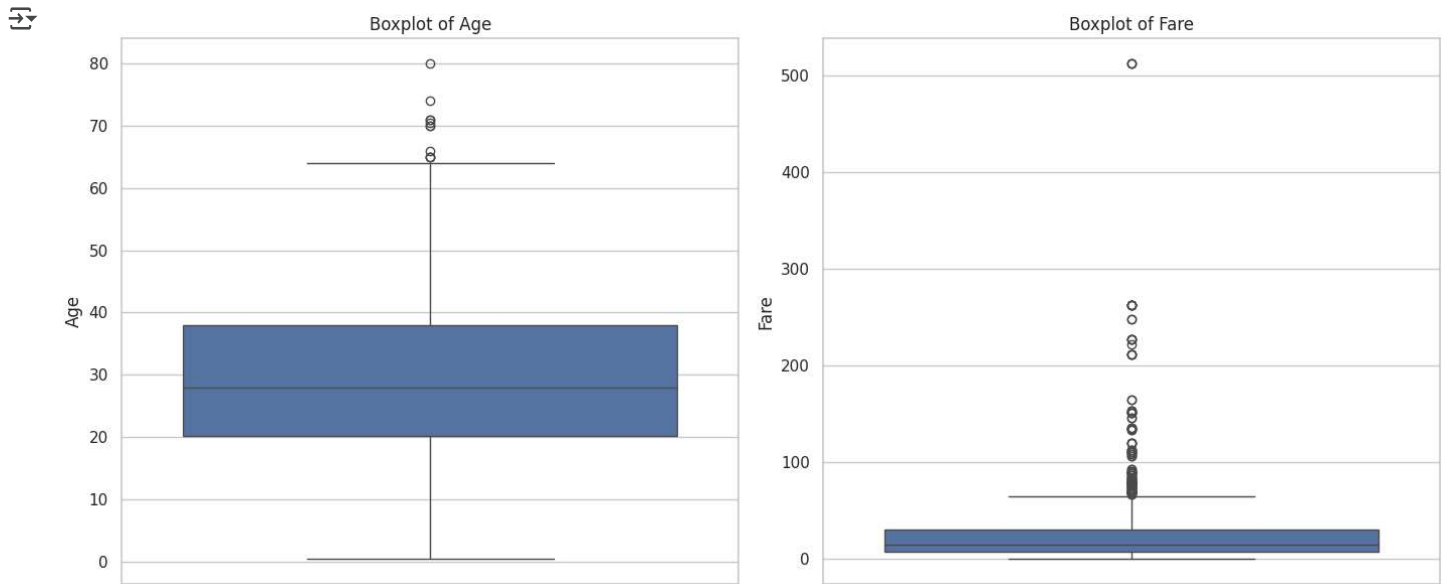
1 # 6. Boxplots to Identify Outliers
2 fig, axes = plt.subplots(1, 2, figsize=(14, 6))

```

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3 sns.boxplot(data=train_df, y='Age', ax=axes[0])
4 axes[0].set_title("Boxplot of Age")
5 sns.boxplot(data=train_df, y='Fare', ax=axes[1])
6 axes[1].set_title("Boxplot of Fare")
7 plt.tight_layout()
8 plt.show()

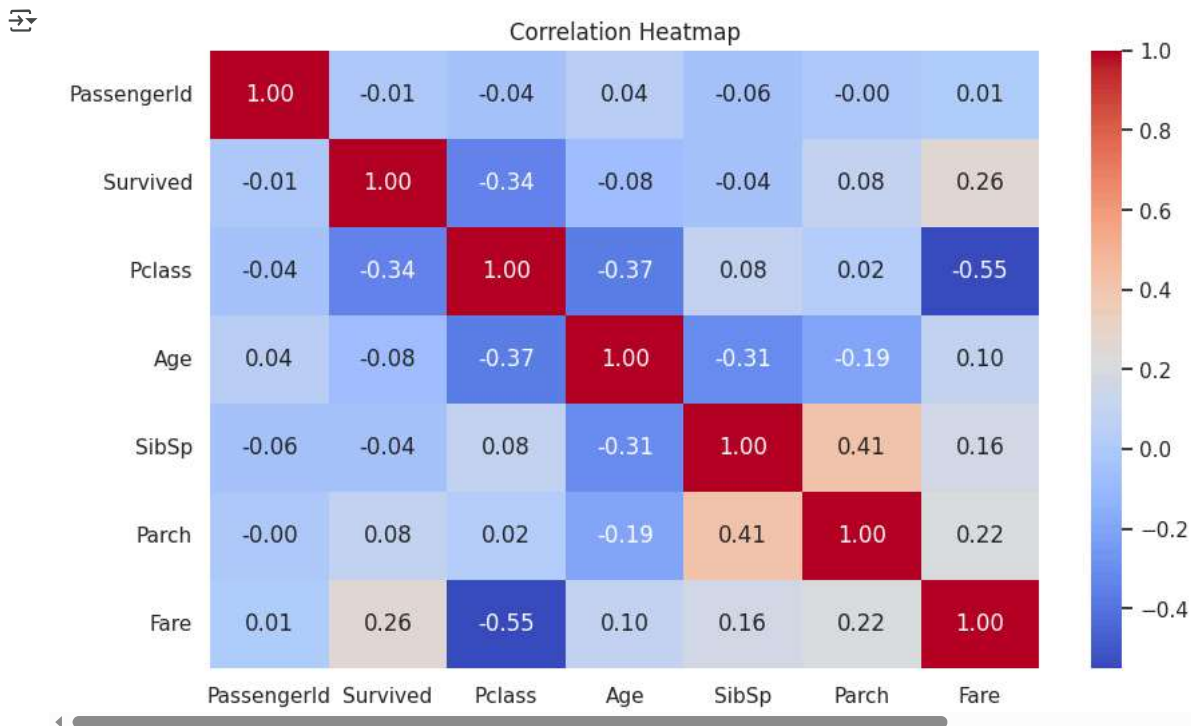
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```

1 # 7. Correlation Heatmap
2 plt.figure(figsize=(10, 6))
3 sns.heatmap(train_df.corr(numeric_only=True), annot=True, cmap='coolwarm', fmt=".2f")
4 plt.title("Correlation Heatmap")
5 plt.show()

```



```

1 # 8. Pairplot
2 sns.pairplot(train_df[['Survived', 'Pclass', 'Age', 'SibSp', 'Parch', 'Fare']], hue='Survived', palette='husl')
3 plt.suptitle("Pairplot of Numerical Features Colored by Survival", y=1.02)
4 plt.show()

```



Pairplot of Numerical Features Colored by Survival



```

1 # 9. Barplots for Categorical Features vs Survival
2 sns.barplot(data=train_df, x='Sex', y='Survived')
3 plt.title('Survival Rate by Sex')
4 plt.show()
5
6 sns.barplot(data=train_df, x='Pclass', y='Survived')
7 plt.title('Survival Rate by Passenger Class')
8 plt.show()
9
10 sns.barplot(data=train_df, x='Embarked', y='Survived')
11 plt.title('Survival Rate by Embarkation Port')
12 plt.show()

```

